

Between Us: An Interactive Installation for Parent-Child Relationship Visualization Using NFC and Generative Algorithms

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Figure 1: Snapshots of the interactive installation *Between Us*.

Abstract

In many East Asian families, early childhood communication breakdowns often lead to long-term parent-child relationship issues. While communication states are inherently invisible, they can be expressed through everyday objects. Inspired by the natural phenomenon of crown shyness, this installation visualizes emotional boundaries within familial ties. Drawing from semiotics, it transforms daily household items into symbolic tokens. Participants assign affective values to twelve NFC-embedded tokens, reflecting subconscious attitudes formed in early interactions. Using Arduino and generative algorithms in TouchDesigner, these values are translated into dynamic canopy forms that depict internal communication states. Combining symbolic visualization, computational aesthetics, and sociocultural reflection, the work offers a new interactive medium for exploring emotional expression and relational understanding in the context of childhood development.

CCS Concepts

• Human-centered computing → Interaction design.

Keywords

Parent-child relationship, Symbolic visualization, Crown shyness, Generative art, Affective interaction

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1 INTRODUCTION

Emotional communication within families plays a vital role in shaping social and psychological development, particularly in collectivist cultures where indirect expression is common. In East Asian contexts, unspoken emotional boundaries formed in childhood can persist into adulthood, yet these boundaries often remain invisible and underexplored. Interactive art and affective computing have offered diverse strategies to visualize emotion, from bodily-based interfaces that surface intimate responses [Schiphorst 2009] to biofeedback-driven installations prompting self-reflection [Goudswaard et al. 2020]. However, few works integrate these approaches with culturally specific metaphors to address parent-child relational dynamics rooted in early experiences. To address this gap, *Between Us* employs the natural phenomenon of crown shyness as a symbolic framework for mapping emotional distance in familial communication. By combining tangible interaction with generative visualization, the work invites reflection on the cultural and emotional architectures that shape childhood development.

2 RELATED WORK

Prior works have explored how tangible interfaces and generative visuals foster emotional self-awareness and meaning-making. For instance, *Mirror Ritual* externalizes inner states through affective machine-generated poetry, encouraging reflection [Rajcic and McCormack 2020]. Similarly, *The Diary of Niels* uses RFID-tagged artifacts with narrative overlays to enhance engagement through symbol-object interaction [Schou and Løvlie 2020]. These works show how physical tokens can mediate complex emotional experiences. Recent work in generative aesthetics has explored multimodal emotion visualization, notably through platforms like TouchDesigner that facilitate real-time generative graphics driven by hardware including LEDs and microcontrollers. Yet few approaches integrate

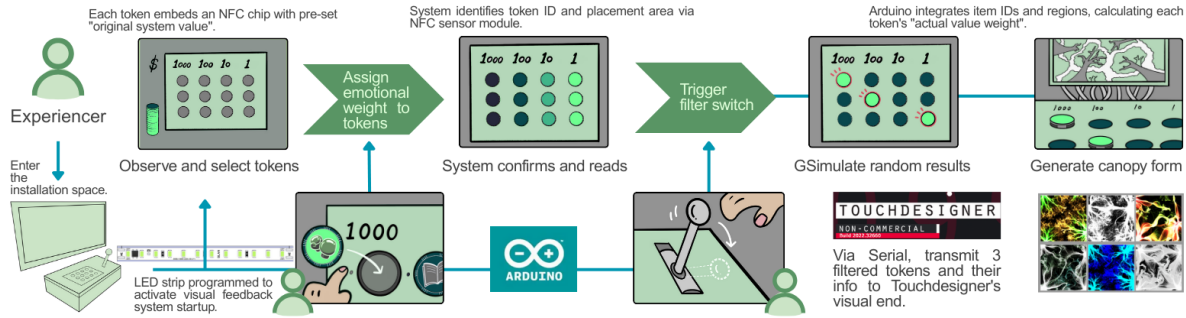


Figure 2: The user interaction flow and technical pipeline of the interactive installation.

generative algorithms with culturally grounded metaphors or address the subtle emotional dynamics of parent-child communication. As a result, they often fail to connect tangible interaction, computational aesthetics, and relational contexts within cultural frameworks. This installation uniquely visualizes emotional boundaries in parent-child relationships through interactive symbolic representation, marking an innovative convergence of semiotics, affective interaction, and computational aesthetics.

3 DESIGN AND IMPLEMENTATION

Between Us is an interactive installation designed to visualize the subtle emotional boundaries within parent-child communication. It invites participants to interact with symbolic tokens representing familiar household objects, prompting reflection on the hidden emotional patterns formed in early life. Through embodied engagement, the work aims to make intangible relational dynamics perceptible and to encourage a deeper awareness of emotional presence in familial contexts.

3.1 Design Concept

Inspired by the natural phenomenon of crown shyness—where neighboring trees leave visible gaps—the work explores unspoken boundaries in relationships. In *Between Us*, familiar objects like mirrors, clocks, and doors become symbolic tokens carrying emotional and cultural meaning. Participants assign personal values to these tokens, reflecting their own parent-child experiences. This process acts as a metaphor for negotiating intimacy, distance, and emotional weight. The installation invites quiet self-reflection through sensory and symbolic interaction rather than explicit storytelling (Figure 1).

3.2 Technical Implementation

The system integrates Arduino-based sensing and NFC interaction with generative visualization in TouchDesigner (Figure 2). The input module consists of twelve NFC-embedded tokens, each assigned an initial weight (1, 10, 100, 1000) representing emotional intensity. Participants place tokens into zones, overlaying default values with personal interpretations. The processing module uses an Arduino to read NFC and placement data, confirmed via LED feedback. A weighted random algorithm then selects three tokens reflecting participants' emotional configurations. The output module sends the data to TouchDesigner, where a custom generative

algorithm produces real-time tree canopy visuals. Variations in density, branching, and spacing metaphorically express relational states from closeness to distance, forming an immersive environment of emotional reflection.

4 EVALUATION

The installation was exhibited at the Art and Exhibition Center of Communication University of China, attracting over 300 participants. Audience feedback indicated that the experience effectively conveyed the significance of emotional boundaries in parent-child relationships, prompting reflection on the importance of emotional presence and communication in family life.

5 CONCLUSION

Between Us bridges artistic expression and computation to visualize the unseen dynamics of parent-child communication. By transforming symbolic tokens into generative, crown-like tree forms, it invites reflection on personal communication patterns. Combining semantic encoding, generative aesthetics, and embodied interaction, the installation advances emotional visualization and intimate interface design. Future work may further explore personalization and data-driven storytelling in family relationship contexts, followed by formal user studies to evaluate its communicative and emotional impact.

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